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CS 600

Q1. No. 26.6.7

(a) Convert the following linear program into standard form:

$$\begin{aligned} \text{minimize:} \quad & z = 3y_1 + 2y_2 + y_3 \\ \text{subject to:} \quad & -3y_1 + y_2 + y_3 \geq 1 \\ & 2y_1 + y_2 - y_3 \geq 2 \\ & y_1, y_2, y_3 \geq 0 \end{aligned}$$

Ans.

Objective (Minimize):
z=3y1+2y2+y3
Subject to constraints (converted to ≤ form):
3y1−y2−y3≤−1
−2y1−y2+y3≤−2
y1,y2,y3≥0

Solution (Using Simplex Method)
The optimal values of the variables are:

- y1=0.2
- y2=1.6
- y3=0

Minimum value of z:
Z = 3(0.2)+2(1.6)+1(0) = 0.6+3.2+0 = **3.8**

Final Answer
Optimal Solution:
y1=0.2, y2=1.6, y3=0
Minimum value of the objective function:
z=3.8

Q2. No. 26.6.26
Give a linear programming formulation to find the minimum spanning tree of a graph. Recall that a spanning tree T of a graph G is a connected acyclic subgraph of G that contains every vertex of G. The minimum spanning tree of a weighted graph G is a spanning tree T of G such that the sum of the edge weights in T is minimized.

Ans.

Linear Programming Formulation for Minimum Spanning Tree
To determine the Minimum Spanning Tree (MST) of a graph, linear programming can be employed. Given a weighted, undirected graph $G=(V,E)G$ where V is the set of vertices and E is the set of edges with associated weights c_e for each edge $e \in E$, the goal is to find an acyclic subgraph $T \subseteq G$ that:

- Contains all the vertices in V,
- Does not contain cycles,
- And minimizes the total edge weight $w(T)=\sum_{e \in T} c_e$

This optimization problem can be approached using linear programming, which models the objective function and constraints using linear relationships. Linear programming is a subset of mathematical programming that seeks to optimize (minimize or maximize) a linear objective function subject to linear equality and inequality constraints.

Variable Definitions:
Let $x_{ij} = 1$ if edge $e=(i,j)$ is included in the spanning tree T, and $x_{ij} = 0$ otherwise.
Let $S \subset V$ represent any nontrivial subset of the vertex set.

Linear Program:
Objective:
Minimize $w(T)=\sum_{e \in E} c_e x_e$
Subject to:

1. Spanning Tree Edge Count: $\sum_{e \in E} x_e = |V|-1$
2. Cycle Prevention (Subtour Elimination):
   For every subset $S \subset V$, where $2 \le |S| \le |V|-1$
   $$\sum_{\substack{e = (i, j) \in E \\ i, j \in S}} x_e \le |S|-1$$
3. Binary Decision Variables:
   $x_e \in \{0,1\} \ \forall e \in E$

Remarks on Mathematical Optimization:
Linear programming is a widely-used optimization framework in which a linear objective function is optimized over a convex polytope—defined as the intersection of finitely many half-spaces (linear inequalities). In this context, Lagrange's inequalities and other convex optimization tools may be used when functions are nonlinear but approximal by linear models. It's important to note that while the terms linear programming and linear optimization are sometimes used interchangeably, they refer to the same concept: optimizing a linear function under linear constraints

Q3. No. 26.6.31

(a) A political candidate has hired you to advise them on how to best spend their advertising budget. The candidate wants a combination of print, radio, and television ads that maximize total impact, subject to budgetary constraints, and available airtime and print space.

type	impact per ad	cost per ad	max ads per week
radio	a	10,000	25
print	b	70,000	7
tv	c	110,000	15

Design and solve a linear program to determine the best combination of ads for the campaign.

Ans.

Linear Programming Formulation for Optimal Ad Allocation

Let:

- x = number of radio ads
- y = number of print ads
- z = number of TV ads
- a, b, and c = impact per ad for radio, print, and TV respectively
- B = total advertising budget

Objective Function:

Maximize: $\text{Impact} = ax + by + cz$

Subject to:

$10{,}000x + 70{,}000y + 110{,}000z \leq B$ (Budget constraint)
$x \leq 25$ (Max radio ads)
$y \leq 7$ (Max print ads)
$z \leq 15$ (Max TV ads)

Non-Negativity Constraints:

$x, y, z \geq 0$